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Project Title:	In Vivo Induction Of Chromosomal Damage: A Spectral Karyotyping Study		
Division Name:	Human Research		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Zip Code:	11794-8691	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	02-OBPR-02
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Flight Assignment:	NOTE: Received NCE to 03/30/2007 per PI (12/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Whorton, Elbert (UTMB at Galveston) Golightly, Marc (Stony Brook University)		
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Task Description:	The new exploration of Moon and Mars will undoubtedly happen in the very near future. Exposure to space radiation during such a journey may create potential health risk to the astronauts for late occurring diseases (such as cancer). Different types of galactic cosmic rays (GCR) are present in space and have large mass and energy (HZE particles). Among these, stripped 56Fe ions are of major concern. Although HZE particles are approximately 1% of GCR, their contribution to health risk could be significant because of their very high linear energy transfer (LET) resulting in a larger amount of energy being deposited in the hit cells. Biological effects of 56Fe ions (measured as cytotoxicity, mutagenicity and chromosomal damage) have been reported but most studies were conducted in in vitro systems using different cell types. Therefore, it is difficult to relate the significance of such in vitro findings to in vivo systems. It has been well established that acute myeloid leukemia is one of the major malignancies associated with exposure to ionizing radiation in human beings and mice. It is therefore considered as one of the most important cancers related to space flights. For radiation-induced genetic damage to be useful for risk assessment, it must be established in in vivo systems especially in cells that are known to be at risk for health problems associated with radiation exposure such as bone marrow cells, the known target cell for radiation-induced leukemia. We exposed male CBA/CaJ mice whole body to varying doses of 1 GeV/amu 56Fe ions or 137Cs gamma rays. We collected bone marrow cells from exposed mice at different times up to 3 months post-exposure. Multiple sampling times permit an accurate determination of relative biological effectiveness (RBE) for 56Fe ion-induced chromosomal damage relative to 137Cs gamma rays, as a function of absorbed dose, aberration type, and time after irradiation. The novel technology for genome-wide multicolor fluorescence in situ hybridization of mouse chromosomes (mFISH) is being used for the detection, analysis, and characterization of chromosome damage in bone marrow cells. Our in vivo study is aimed to address the following hypotheses: (1) Complex chromosomal exchanges are more prevalent in the BM cells of mice exposed to 56Fe ions than those exposed to similar doses of 137Cs gamma rays. (2) There is a dose-dependent increase in the frequency of chromosome aberrations in bone marrow cells of mice exposed to 56Fe ions or 137Cs gamma rays. (3) Genomic instability (expressed as delayed appearance of chromosome aberrations) and clonal expansion of chromosomal exchanges commonly found in mice exposed to 56Fe ions and 137Cs gamma rays are detected in a fraction of surviving cells at later times post-irradiation. The result from this study will provide pivotal information that should lead to a better understanding of dose-response kinetics and of mechanisms by which 56Fe ions induced aberrations can potentially be involved in the induction of genomic instability and cancer. The result also would be important in improving risk assessment of protracted space flight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Upon completion of our in vivo work, the data will definitely provide pivotal information that should lead to a better understanding of dose-response kinetics and of mechanisms by which 56Fe ions induced chromosome aberrations can be potentially involved in the induction of genomic instability and cancer. This statement is based upon the certainty that chromosome aberrations have been proven to be the most relevant endpoint for detecting carcinogenic activity of suspect agents because of their strong correlation with the induction of cancers. The results also would be important in improving risk assessment of protracted space flight. Both 137Cs gamma rays (the reference radiation) and 56Fe ions are used in this study. The results would therefore facilitate determining whether radiation quality influences the probabilities for increased carcinogenesis from space radiation. Moreover, chromosomal damage is measured at the low-dose level of 56Fe ions (i.e. 0.1Gy). Consequently, the results should allow us to make a meaningful comparison for understanding the in vivo RBE of 56Fe ions on the target cells as a function of absorbed dose, aberration type and time. The results also would lead to a better understanding of basic mechanisms of cytogenetic damage potentially involved in cancer induction by these two different sources of radiation.
Task Progress:	Analysis of metaphase chromosome aberrations in mouse bone marrow cells by means of the whole genome multicolor fluorescence in situ hybridization (mFISH) We currently have completed the evaluation of dose-response relationships for the induction of several types of chromosome aberrations in metaphases prepared from bone marrow cells collected at day 7 from male CBA/CA mice exposed to different doses of 1 GeV/amu 56Fe ions (0, 0.1, 0.5 and 1.0 Gy) or 137Cs gamma rays (0, 0.5, 1.0 and 3.0 Gy, at the dose rate of 0.72 Gy/min using a GammaCell40 located at Stony Brook University). At day 7 post-irradiation, we detected all types of aberrations in bone marrow cells collected from mice exposed to either 56Fe ions or 137Cs gamma rays. These were chromosomal exchanges and breaks (both chromatid- and chromosome- types). Chromosomal exchanges included translocations (robertsonian or centric fusion, reciprocal and incomplete types) and dicentrics. In general, we found no evidence of a non-random involvement of specific chromosomes in any type of aberrations observed in mice exposed to 56Fe ions or 137Cs gamma rays. We also found that the highest dose of 56Fe ions (1.0 Gy) induced more cells with multiple aberrations, as compared to those induced by the high dose of 137Cs gamma rays (3.0 Gy). At the radiation dose range used in our in vivo study, the majority of chromosomal exchanges were simple. Among the observed translocations, the frequencies of reciprocal translocations were less than that of the incomplete type. Although exchanges between homologous chromosomes (e.g. dicentrics or robertsonian translocations) were detected, the non-homologous exchanges were more prevalent. We also detected complex exchanges but these were found in bone marrow cells collected from mice exposed to high doses of radiation (i.e 1 Gy of 56Fe ions or 3 Gy of 137Cs gamma rays) only. However, the frequencies of complex exchanges were low, i.e. 4% of total observed exchanges found in 56Fe-ion-exposed mice and 3% of those found in 137Cs gamma-ray-exposed mice. Overall, our in vivo data did not indicate that complex chromosome rearrangements were the signatures of exposure to 56Fe ions. Each type of radiation induced significant dose-dependent increases (ANOVA, $p<0.01$) in the frequencies of chromosomal damage, including the numbers of abnormal cells. We used both linear and linear-quadratic regression equations to test the fitness of the data, i.e. percent abnormal cells, numbers of breaks (both chromatid and chromosome types), and numbers of exchanges (all types). However, the use of a linear–quardratic regression equation did not improve the fit of the data, so the simpler linear equation was used to describe the dose-response curves. Based upon the

	slopes of the dose-response curves (linear equation), 56Fe ions were 1.7 to 5 times more effective than 137Cs gamma rays in inducing chromosomal damage (depending on the type of damage). Using the mouse in vivo micronucleus assay, we previously observed similar results (last year Task Book). Overall, our in vivo data indicate that, at a similar dose, 56Fe ions are more effective in inducing chromosomal damage in bone marrow cells of exposed mice than 137Cs gamma rays. The results could have important implications for the development of a better policy on risk assessment for space flight.
Bibliography Type:	Description: (Last Updated: 01/20/2026)
Abstracts for Journals and Proceedings	Rithidech K, Golightly M. "Analysis of Cell Cycle in Mouse Bone Marrow Cells Following an Acute in vivo Exposure to 56Fe Ions." 15th Annual NASA Space Radiation Health Investigators' Workshop, Port Jefferson, New York, May 16-20, 2004. Proceedings, 15th Annual NASA Space Radiation Health Investigators' Workshop, May 2004. , May-2004
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Presentation	Supanpaiboon, W.; Honikel, L.; Brooks, A.L.; Rithidech, K. " In vivo induction of chromosomal damage" the 16th Annual NASA Space Radiation Investigation Workshop May-2005
Presentation	Rithidech, K.; Honikel, L.; Whorton, E.B. "Analysis of chromosomal damage after in vivo exposure to 56Fe ions by means of mFISH and micronucleus methods" COSPAR Colloquium Jul-2006
Presentation	Rithidech, K.; Honikel, L.; Whorton, E.B. "mFISH analysis of chromosomal damage in bone marrow cells collected from CBA/CaJ mice following whole body exposure to 56Fe ions" the 4th Intl and the 17th Annual NASA Space Radiation Investigators' Workshop Jun-2006